Reconfigurable Modular Null-Steering / CP Antenna — Designer & Verification Guide

Companion to: K. M. Younus, K. Sultan, C. Fumeaux, A. Abbosh, “Reconfigurable Modular Antenna System With Null Steering and Circular Polarization”, IEEE Trans. Antennas Propag., vol. 73, no. 5, May 2025.

This guide is laid out the way a design review would go: what the system is, what each claim rests on, what the paper leaves out, and how to verify every block in HFSS, MATLAB and Python before building hardware.

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1. The system in one page

One unit = phase shifter (PS) + printed dipole on one 25 × 60 mm board. - **Phase shifter:** two elliptical patches on the outer layers, broadside-coupled through an elliptical slot in the inner ground. Each patch’s free end has a stub, a DC-block capacitor and a varactor to ground, and one bias voltage tunes both varactors. The quoted performance is 145° tuning range, ±2° imbalance and IL < 1 dB over 5–7 GHz. - **Antenna:** a wideband printed dipole with an integrated microstrip-to-slotline balun. It fires endfire (off the board tip) with a broad beam of about 5 dBi. - **Built-in 180°:** a mirror-image board (type B) radiates in antiphase with type A for the same feed.

Systems: units fed by an equal-split power divider.

System	Layout	Claimed performance
1-D	2 units coplanar (A, B), spacing d = 25 mm = λ0/2 at 6 GHz	The antiphase pair has a null at broadside; tuning the two PSs steers it to ±40° with > 15 dB depth
2-D	2 × 2 units (A, B / B, A); every x-pair and y-pair is antiphase	Null planes at u = u0 and v = v0; ±40° per plane with > 10 dB depth
CP	4 units as a cross (X), 90° set on one orthogonal pair	RHCP or LHCP, AR < 2.3 dB over 5–7 GHz

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2. Terminology (how an RF designer describes this design)

Term	Meaning here
Reflection-type phase shifter (RTPS)	What the PS really is. A quadrature (3-dB backward) coupler whose through and coupled ports see identical reflective loads Γ. Then S21 = 2ΓTK and S11 = Γ(T² + K²), which are paper eqs. 6–7. The phase of S21 follows the phase of Γ.
Broadside-coupled coupler	Two strips/patches on opposite sides of a thin dielectric, coupled through an aperture in the ground between them. It gives tight (3 dB), wideband coupling in a small area (Abbosh, ref. [30]).
Even/odd-mode analysis, Z0e/Z0o, coupling factor C	Standard coupled-line analysis behind eqs. 1–2. C = (Z0e − Z0o)/(Z0e + Z0o); 3 dB means C = 0.707.
Reflective load / transformer stub	The ~λg/4 stub transforms the varactor impedance. This widens the reflection-phase swing (Fig. 3).
Varactor (hyperabrupt GaAs)	MAVR-000120-14110P, Ci 0.14–1.7 pF (12:1 ratio), Rp ≈ 2 Ω. It is reverse-biased: cathode to the bias node, anode to ground.
Bias tee / RF choke / DC block	Choke (paper 100 nH) feeds DC to the varactor while blocking RF. The DC block (paper 39 pF) keeps bias off the RF path.
Self-resonant frequency (SRF)	Above its SRF a real inductor looks capacitive and a real capacitor looks inductive. A 0402 100 nH part has SRF ≈ 1–2 GHz and 39 pF ≈ 2 GHz, both far below 5–7 GHz (section 9).
Phase imbalance / in-band phase error	Deviation of a state's phase shift across 5–7 GHz from its band-average.
Microstrip-to-slotline balun	The back-side microstrip crosses the front-side slot line and ends in an open stub. The slot ends in a square cavity (open). Together they convert unbalanced feed to balanced dipole drive.
Endfire printed dipole	Arms beyond the ground edge; the ground acts as a reflector, so the beam points off the board tip (+y board, +z world).
Decoupling stub	The ls strip on the ground edge between neighbouring antennas, which reduces mutual coupling (paper "stub between the two antennas").
Antiphase pair / difference pattern	Two identical elements fed 180° apart give a broad pattern with a null broadside to the pair axis.
Null steering	Moving that null by a differential phase Δφ = k·d·sinθ_null.
Null depth	Pattern level at the null relative to the beam peak in the same cut.
Rejection at the interferer angle	What matters in practice: the level *at the jammer's direction*, not the deepest point of the null anywhere.
uv-space	u = sinθ·cosφ, v = sinθ·sinφ. The 2-D nulls are straight lines u = u0 and v = v0 (Fig. 12).
Beam / null squint	A phase shifter is not a true-time delay, so a fixed phase setting gives a null angle that changes with frequency.
Axial ratio (AR), LHCP / RHCP	CP purity and sense. They come from two orthogonal linear components in quadrature (90°).
Crossed-dipole (turnstile) CP	Two orthogonal dipole pairs fed 90° apart. Here each "dipole" is a pair of opposite wings.
Embedded element pattern	One unit's pattern measured or simulated inside the array with the others terminated. It includes mutual coupling and is what you feed into the array model for verification.
Phase-matched cables, splitter imbalance	External feed errors that add to the PS errors (ZX10-2-183-S+ 2-way, ZN4PD-5R183-S+ 4-way in the paper).
Bias look-up table (LUT)	Mapping from wanted null angle and frequency to phase, then capacitance, then voltage, per unit.

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3. Verification of the paper's claims (done here)

Claim	Check	Result
PS 145–150° range (analytic)	Eqs. 3–7 with paper values (Zcl = Zstub = 50 Ω, 3-dB coupler, Lp = 0, ideal 100 nH / 39 pF)	**114–116°.** For any single-varactor RTPS the limit is 4·atan(√r) − 180° = **116°** for r = 1.7/0.14. 145° needs ≈ 0.3–0.4 nH series inductance (package + pads), which gives 136–147° (`rtps_model.py` , `ns_demo_phase_shifter.m`).
±2° imbalance	Same model, worst state deviation over 5–7 GHz	±4° (ideal) to ±6–8° with the inductance that gives 145°. Each source defines imbalance differently, so compare like with like in HFSS.
IL < 1 dB	Rp = 2 Ω alone	≈ 0.65–0.70 dB worst state. Substrate and conductor loss come on top, so the measured 0.8 ± 0.15 dB is plausible.
±40° 1-D steering	Δφ = k·d·sin40° = 116° at 6 GHz, split ±58° over the two PSs	Feasible. Max steering ±52° at 6 GHz, ±44° at 7 GHz, ±65° at 5 GHz. **Both** PSs must be tuned; one alone gives only 0–145°, i.e. a one-sided null.
Null over 5–7 GHz	Fixed bias set for 40° at 6 GHz	**Squint:** the null sits at 50.5° at 5 GHz and 33.4° at 7 GHz. The bias LUT must be frequency-dependent (`LUT_1D_null_angle_to_bias.csv`).
> 15 dB null depth	Monte Carlo, 0.5 dB / 4° rms per channel	Rejection at the interferer angle: median 26 dB, 10th percentile 20 dB, null pointing error under 3.5° (90th pct). **Amplitude error fills the null; phase error moves it.**
2-D ±40° in both planes	Element phases ±αx ± αy must fit in the PS range	Envelope **|u0| + |v0| ≤ 0.79** at 6 GHz. ±40° in one plane is fine, but simultaneous steering is limited (e.g. about 23.6° in both).
CP with Fig. 1(c) colours (U1, U4 red; U2, U3 blue)	Array model, all wings mounted the same way (board x radially outward)	**Opposite wings cancel at broadside**, giving a null instead of CP. Opposite wings must radiate co-phased, i.e. be opposite types (A / B), giving AR = 0 dB ideal. See `SYS_CP_cophased` in the placement table. Verify the sense mapping (which pair gets +90° for RHCP) in HFSS; it depends on numbering and mounting.
AR < 2.3 dB	AR vs quadrature / amplitude error	10° quadrature error gives 1.5 dB; plus 1 dB amplitude imbalance gives 1.8 dB. Consistent with the measurement.

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4. Geometry, stack-up and what had to be assumed

Stack-up (unit and PS boards):

Layer	z (mm)	Contents
TOP (front)	1.116	PS input feeder, top patch, top stub chain; dipole arms and ground
RO4003C	0.608–1.116	εr 3.55, tanδ 0.0027, 0.508 mm
MID (GND)	0.608	PS ground with elliptical slot
RO4450F prepreg	0.508–0.608	εr 3.52, tanδ 0.004, 0.1 mm
RO4003C	0–0.508	0.508 mm
BOT (back)	0	PS output feeder, bottom patch, bottom stub chain; dipole feed loop

The 50 Ω widths are 1.145 mm on TOP (0.508 mm to MID) and 1.37 mm on BOT (0.608 mm to MID).

Dimensions used from the paper: - **Phase shifter (Fig. 2):** L 36, W 25, Fl 13.7, Pl 8.8 (listed as “l”), Pw 4.9, Sl 8.8, lstub 9.4, Stl 6.1, h 0.508, prepreg 0.1. - **Stub chain:** Stl = 6.1 mm is the line from the patch to the DC block; the DC block plus junction pad brings the total to the varactor to lstub = 9.4 mm. This reading makes the top (Stl1) and bottom (Stl2) chains identical, as the paper’s symmetry statement requires. - **Dipole (Fig. 6):** Lant 24, W 25, w1 1.6, w2 1.07, l1 1.65, l2 3.1, l3 14.2, l4 4.5, l5 4, l6 4.1 (printed as “li”), l7 0.43, gs 0.11.

Item	Problem in the paper	What this pack does
Dipole substrate	Text says “RT/Duroid 4003C εr = 2.2”. RO4003C is εr 3.55; 2.2 is RT/duroid 5880.	εr 3.55. Supporting evidence: w2 = 1.07 mm is 50 Ω on 0.508 mm εr 3.55, but would be ≈ 64 Ω on εr 2.2.
Slot Sw	Listed Sw = Sl = 8.8 (a circle), but Fig. 2(e) shows an ellipse with Sw/Sl ≈ 0.6	Both variants: `PS_slot-paper-8p8x8p8` and `PS_slot-drawing-ratio-8p8x5p3` . Simulate both; the one giving 3-dB coupling at 6 GHz is the intended one.
l6 / ls	Both appear in Fig. 6; only one “li = 4.1” is listed. The ls strip width is not given.	l6 = 4.1, ls = 4.1, ls width 1.2 [ASSUMED]. ls is the decoupling stub, so tune it for minimum S21 between neighbouring units.
Fig. 6 scale	The drawing is not to scale (its vertical scale differs from the horizontal)	Dimensions taken from the list, never from the drawing.
Feed-loop open end	Tail length not stated	Right leg = l5 − l7. Tune for S11 < −10 dB over 5–7 GHz.
MID ground under the antenna	Fig. 8 shows the MID ground ending at the dash-dot line (y = 36). The dipole then sits on the full 1.116 mm stack with no inner ground, so the back feed line is referenced to the front ground 1.116 mm away (≈ 80 Ω for 1.07 mm).	Two unit variants: `mid-paper` (as drawn) and `mid-extended` (MID continues under the dipole ground with the same square slot and slot line, stitched by vias). The extended version keeps the 1.07 mm feed at ≈ 58 Ω. The dipole must be re-tuned on the 3-metal stack either way, because the 0.508 mm standalone dimensions do not carry over unchanged.
Parts	100 nH / 39 pF / varactor footprints not given	0402-class footprints. Check the MACOM land pattern for “-14110P”.
Vias	Count and diameter not given	0.3 mm, 3 per connector side, 6 per DC pad. **Through-vias are feasible:** every through-via conflict lands on ground copper (checked), so no blind vias or sequential lamination are needed.

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5. Boards in this pack

All boards include silkscreen with refdes, port labels, varactor polarity bar, bias labels, unit number, type A/B, beam direction and polarisation arrow. The `<ORG> <PROJECT> REV YYYY-WW` field is a placeholder for your own marking.

Folder	Board	Purpose
A_without_antenna	PS standalone, 3 versions (slot per text, slot per drawing, resistor bias)	2-port VNA characterisation: S21 phase vs bias, IL, RL; builds the LUT.
B_antenna_only	Dipole on 0.508 mm, 2 layers	S11 and pattern of the bare antenna (paper Fig. 7).
C_units_with_antenna	Unit A / B × MID paper / extended	The integrated element (Fig. 8).
D_system_1D_2units	Coplanar panel U1 (A) + U2 (B), plus top-view assembly drawing	1-D null steering (Fig. 8).
E_system_2D_4units	Two coplanar rows (front: U4 A, U3 B; back: U2 B, U1 A) 25 mm apart, plus assembly drawing	2-D null steering (Fig. 11).
F_system_CP_4units	Four wing boards, plus assembly drawings for both type assignments	CP (Fig. 13).
G_feed_dividers	Wilkinson 1:2 and 1:4, 6 GHz, 25 mm output pitch, 100 Ω 0402	Integrated, phase-equal replacement for the external splitters.
H_fab_panel	U1–U4 + PS coupon + dipole coupon, fiducials	One fabrication order covering all builds and test coupons.

Supporting files: - `assembly_3D_placement.csv` : world position and azimuth of every board for each system. - `BOM_and_placement.csv` : refdes, value, side, x/y, polarity. - `DRC_report.txt` : copper gaps, thin features, silkscreen over copper.

DXF layers:

Layer	Contents
OUTLINE	board edge
TOP / MID / BOT	merged copper
*_CUTOUT	holes, to subtract in HFSS
VIA	drill circles
PORT	port reference
SILK_TOP / SILK_BOT	silkscreen
ASSY_TOP / ASSY_BOT	component bodies
FAB	stack-up notes

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6. Ports, feeds and bias: with and without antenna

Test	Port definition
PS alone	J1 edge SMA on TOP at y = 0 (input), J2 edge SMA on BOT at y = 36 (output). In HFSS: lumped ports, a 0.508 mm (TOP–MID) and a 0.608 mm (BOT–MID) sheet. Prefer wave ports with a de-embedded launch for final numbers.
Dipole alone	Edge SMA at y = 0: centre pin on the BOT feeder, body on the front ground.
Unit	Only J1 (TOP, y = 0). The PS output continues directly into the dipole feed on BOT.
1-D	U1 and U2 inputs from a 2-way divider through phase-matched cables (or the Wilkinson board). Two bias voltages, one per unit.
2-D / CP	4-way divider, four bias voltages.

Bias wiring: - Each unit has VB1 (TOP) and VB2 (BOT) at the same board edge. Tie them together: both varactors of a PS must see the same voltage, which is the paper's "single bias per phase shifter". - Add 100 pF + 100 nF decoupling at the connector. - Route bias wires along the board edge, perpendicular to the E-plane and away from the antennas. Long wires near a dipole re-radiate and fill nulls; resistive or ferrite-loaded wires are best.

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7. HFSS workflow

1. **Import.** Import the DXF and map TOP to z = 1.116, MID to z = 0.608 and BOT to z = 0. Subtract each *_CUTOUT from its layer, assign PerfE (finite-conductivity Cu in the final run), and add the three dielectric boxes. Model vias as PEC cylinders.
2. **Components as lumped RLC sheets** across the pad gaps (positions in `BOM_and_placement.csv`):

Part	Model
Varactor	Series Rp = 2 Ω, Lp = 0.35 nH [assumed], C = variable `Cv`
DC block	Series 39 pF + 0.4 nH
Choke	Parallel 100 nH ‖ 0.11 pF (SRF 1.5 GHz) or the vendor S2P; alternatively the recommended 10 kΩ

   Better still, import the vendor S2P/SPICE models via Circuit co-simulation.
3. **Phase shifter.** Parametric sweep of `Cv` from 0.14 to 1.7 pF (both varactors equal) over 5–7 GHz. Extract ∠S21 relative to the 1.7 pF state, |S21| and |S11|. Compare with `rtps_cases.png` and the paper's Fig. 5.
4. **Dipole.** Air box with ≥ λ/4 at 5 GHz (15 mm) of clearance, radiation boundary (or PML), 5–7 GHz sweep. Check S11 < −10 dB and broad endfire beam around 5 dBi (paper Fig. 7).
5. **Systems.** Place the boards from `assembly_3D_placement.csv`, or run `04_hfss/hfss_ns_build.py 1D|2D|CP|CPcophased`. Drive all ports with 1 W at 0° (an ideal splitter). Set each unit's `Cv` from the LUT.
6. **Verification sequence:**
   1. Both PSs at mid-range should put the null at broadside.
   2. Step to the LUT values for ±25° and ±40° at 5, 6 and 7 GHz.
   3. Read the null angle, depth and gain (paper Fig. 10).
   4. 2-D: plot the uv map (paper Fig. 12).
   5. CP: first excite **one opposite pair only** and confirm co-phasing at broadside, then add the 90° pair and read AR and sense (paper Figs. 14–15).

7. **Embedded patterns.** Export each unit's embedded pattern (others terminated) to CSV. `array_models.load_element_csv()` replaces the analytic element, so the Python and MATLAB array results become verification-grade, including mutual coupling.

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8. Python and MATLAB

File	What it does
`02_python/rtps_model.py`	Eqs. 1–7 RTPS model, varactor C(V), bias-network parasitics (cases A–D), phase-to-C-to-V LUT. Writes `rtps_cases.png`.
`02_python/array_models.py`	1-D / 2-D / CP systems, steering limits, squint, uv maps, CP co-phasing check, AR sensitivity, Monte Carlo. Writes results/*.png and* .csv.
`02_python/nsgeom.py`, `build_boards.py`	Single geometry source; regenerates every DXF, preview, BOM, DRC and assembly file.
`03_matlab/ns_demo_phase_shifter.m`	MATLAB copy of the RTPS study. Numbers identical to Python (checked in Octave 8.4).
`03_matlab/ns_demo_system.m`	MATLAB copy of the system study: null placement, squint, 2-D envelope, CP check, Monte Carlo.
`04_hfss/hfss_ns_build.py`	PyAEDT build of PS / dipole / unit / 1-D / 2-D / CP with variables per varactor. Not executed here (no licence).

Equation map:

Paper	Python	MATLAB
eqs. 1–2	`coupler_TK`	inside `ns_rtps`
eq. 3	`load_gamma`	`ns_rtps` (G)
eq. 4	`tline_zin`	`ns_rtps` (Zin)
eq. 5	`Z_varactor`	`ns_rtps` (Zv)
eqs. 6–7	`rtps_S`	`ns_rtps`
Δφ = k·d·sinθ (1-D)	`phases_1d`	`ns_phases_1d`
2-D	`phases_2d`	`ns_phases_2d`
AR	`axial_ratio_db`	`ns_axial_ratio`

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9. Important design steps the paper does not cover

1. **Component SRF.**
   - **Choke:** a 0402 100 nH part resonates at about 1–2 GHz, so at 5–7 GHz it is a small capacitor (≈ 0.1 pF) across the varactor. That reduces the effective C ratio.
   - **DC block:** 39 pF resonates near 2 GHz, so at 6 GHz it looks like +14 Ω of series inductance. Together with the pads this is what makes 145° possible at all (case C).
   - **Recommendation:** co-simulate with the vendor S-parameter files, or bias through a 10 kΩ resistor (the varactor draws only nA). A resistor has no SRF problem; it increases the range in the model (case D) but also the imbalance. Choose by simulation.
2. **The varactor C(V) curve and its spread.** Use the datasheet table, not the fitted model in this pack, and measure every PS coupon. The LUT is per unit, per frequency.
3. **Temperature.** Varactor capacitance drifts with temperature, so the null angle and depth drift too. Include a calibration or temperature-compensation step if the null must hold over temperature.
4. **Linearity under interference.** The interferer you want to null passes through each phase shifter at full power *before* the cancellation at the combiner. Strong signals self-bias the varactors, causing phase modulation and IMD that fill the null. Check P1dB / IIP3 of the PS at the expected interferer level; back-to-back varactors are the usual fix.
5. **Amplitude vs state.** The PS loss changes by up to ≈ 0.6 dB with state. Mismatched states in the two arms fill the null (≈ 0.5 dB rms caps rejection near 25 dB). Choose LUT pairs that also equalise |S21|, or add an amplitude trim.
6. **Frequency-dependent LUT.** The null squints by ±8° over 5–7 GHz for a fixed bias.
7. **2-D steering envelope.** |u0| + |v0| ≤ 0.79 with a 143° PS. State this as the specification, not "±40° in both planes".
8. **CP co-phasing.** Opposite wings must add at broadside; check single pairs first and confirm the sense mapping.
9. **External feed errors.** Splitter imbalance (typically ±0.3 dB / ±3°) and cable phase add directly to the null error budget. Use phase-matched cables (or the Wilkinson boards in G_feed_dividers) and calibrate them out.
10. **Mutual coupling and decoupling stub.** Edge units have different embedded patterns. Verify with embedded patterns, not the isolated element.
11. **Stack-up reality.** RO4003C design Dk is 3.55, while the process (clamped stripline) value is 3.38; the prepreg adds about 0.1 mm. The slot line gs = 0.11 mm is at the limit of standard etching, so specify ±0.02 mm or widen it and re-

tune.

12. **MID ground termination** under the antenna and the feed-reference transition at y = 36 (section 4).
13. **Bias-line radiation, ESD, decoupling** (section 6).
14. **Measurement.**
    - Null-depth measurement needs ≥ 40 dB chamber dynamic range and ≤ 0.5° angular resolution near the null.
    - Use uv raster scans for 2-D.
    - Measure CP with a rotating linear source or dual-pol probe.
    - Measure realized gain with a reference horn (paper: 4.6–8.1 dBi).

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10. Researcher's verification checklist

Step	Pass criterion
1. HFSS: PS alone, both slot variants, Cv sweep	Coupler ≈ 3 dB at 6 GHz; range ≥ 140°; IL ≤ 1 dB; RL ≥ 15 dB over 5–7 GHz
2. Fab + VNA: PS coupon	∠S21(V) table at 5/6/7 GHz → per-unit LUT
3. HFSS: dipole alone (0.508 mm), then inside the unit (both MID variants)	S11 < −10 dB over 5–7 GHz; endfire gain ≈ 5 dBi
4. HFSS: 1-D with LUT states	Null at 0, ±25, ±40° ± 3°; depth ≥ 15 dB at 5/6/7 GHz
5. HFSS: 2-D	Null planes in uv; depth ≥ 10 dB; states inside the |u0| + |v0| envelope
6. HFSS: CP (co-phased)	AR ≤ 2.3 dB at broadside over 5–7 GHz; both senses
7. Python/MATLAB with HFSS embedded patterns	Network model agrees with full-wave within 1 dB / 2°
8. Chamber	Patterns vs simulation; null pointing ≤ 3°; realized gain 4.6–8.1 dBi

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11. Paper targets

Item	Paper
PS	145° range (sim), ±2° imbalance, IL 0.8 ± 0.15 dB, RL > 20 dB (analytic), 5–7 GHz (33.3%)
Dipole	S11 < −10 dB 5–7 GHz, realized gain ≈ 5 dB
1-D	±40°, depth > 15 dB, gain 4.6–7.8 dBi, sim/meas null shift < ±3°
2-D	±40° in xz and yz, depth > 10 dB, gain 6.1–8.1 dBi
CP	AR < 2.3 dB (meas.) 5–7 GHz, LHCP and RHCP
Varactor biasing	1-D: 4 varactors, 2 voltages; 2-D / CP: 8 varactors, 4 voltages